

Experiment no. 5

Name: Urvi Shivalkar

PRN: B25ET1196

Class: SY-3(C)

Title: Implement Stack and Queue using arrays.

Code: Implement stack using array

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define MAX 5
```

```
int stack[MAX];
```

```
int top = -1;
```

```
void push(int value) {  
    if (top == MAX - 1) {  
        printf("\nStack Overflow! Cannot push %d\n", value);  
    } else {  
        top++;  
        stack[top] = value;  
        printf("\nSuccessfully pushed %d into the stack.\n", value);  
    }  
}
```

```
void pop() {  
    if (top == -1) {  
        printf("\nStack Underflow! The stack is already empty.\n");  
    } else {  
        printf("\nPopped element: %d\n", stack[top]);  
        top--;  
    }  
}
```

```

void display() {
    if (top == -1) {
        printf("\nThe stack is empty.\n");
        printf("Current TOP position: -1\n");
    } else {
        printf("\nCurrent TOP position: %d\n", top);
        printf("Stack elements (Top to Bottom):\n");
        for (int i = top; i >= 0; i--) {
            printf("| %d |\n", stack[i]);
        }
    }
}

int main() {
    int choice, value;

    do {
        printf("\n--- STACK OPERATIONS MENU ---\n");
        printf("1. PUSH\n");
        printf("2. POP\n");
        printf("3. DISPLAY\n");
        printf("4. DISPLAY TOP POSITION\n");
        printf("5. EXIT\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter value to push: ");
                scanf("%d", &value);
                push(value);
                break;
            case 2:
                pop();
                break;
            case 3:

```

```

        display();
        break;
    case 4:
        printf("\nCurrent TOP position is: %d\n", top);
        break;
    case 5:
        printf("\nExiting program.\n");
        break;
    default:
        printf("\nInvalid choice! Please enter a number between 1 and 5.\n");
    }
} while (choice != 5);

return 0;
}

```

Output:

```

--- STACK OPERATIONS MENU ---
1. PUSH
2. POP
3. DISPLAY
4. DISPLAY TOP POSITION
5. EXIT
Enter your choice: 1
Enter value to push: 4

Successfully pushed 4 into the stack.

```

```

--- STACK OPERATIONS MENU ---
1. PUSH
2. POP
3. DISPLAY
4. DISPLAY TOP POSITION
5. EXIT
Enter your choice: 2

Popped element: 4

```

```

--- STACK OPERATIONS MENU ---
1. PUSH
2. POP
3. DISPLAY
4. DISPLAY TOP POSITION
5. EXIT
Enter your choice: 3

The stack is empty.
Current TOP position: -1

```

```

--- STACK OPERATIONS MENU ---
1. PUSH
2. POP
3. DISPLAY
4. DISPLAY TOP POSITION
5. EXIT
Enter your choice: 5

Exiting program.

```

Code: Implement queue using array

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
#define size 5
struct Queue{
    int que[size];
    int front ,rear;
}Q;
int Qfull(){
    if(Q.rear>=size-1){
        return 1;
    }
    else{
        return 0;
    }
}
int insert(int item){
    if(Q.front==-1){
        Q.front = 0;
    }

    Q.que[++Q.rear]=item;
    return Q.rear;
}
int Qempty(){
    if((Q.front==-1||Q.front>Q.rear))
        return 1;
    else
        return 0;
```

```

}
int delet(){
    int item;
    item = Q.que[Q.front];
    Q.front++;
    printf("\n Delete item is %d" , item);
    return Q.front;
}
void display(){
    int i;
    printf("The element are:\n");
    for(int i = Q.front;i<=Q.rear;i++){
        printf("%d " , Q.que[i]);
    }
}
void main(){
    int choice,item;
    char ans;
    Q.front = -1;
    Q.rear = -1;
    do{
        printf("\n Main menu");
        printf("\n1.Insert\n2.Delet\n3.Display");
        printf("\nEnter your choice:");
        scanf("%d" , &choice);
        switch(choice){
            case 1:
                if(Qfull()){
                    printf("\n Queue is full");
                }
            else{

```

```

        printf("Enter the element to insert:");
        scanf("%d" , &item);
        insert(item);
        break;
    }
    case 2:
    if(Qempty()){
        printf("Queue is empty:");
    }
    else{
        item = delet();
        break;
    }
    case 3:
    if(Qempty()){
        printf("\n Queue is empty");
    }
    else{
        display();
    }
    break;

    default:
        printf("\n wrong choice");
        break;
    }
    printf("\n Do you want to continue?");
    ans = getch();
} while(ans=='Y' || ans=='y');
}

```

Output:

```
Main Menu
1. Insert
2. Delete
3. Display
Enter your choice: 1
Enter the element to insert: 1

Inserted item is 1
Do you want to continue? (y/n): y
```

```
Main Menu
1. Insert
2. Delete
3. Display
Enter your choice: 1
Enter the element to insert: 2

Inserted item is 2
Do you want to continue? (y/n): y
```

```
Main Menu
1. Insert
2. Delete
3. Display
Enter your choice: 1
Enter the element to insert: 3

Inserted item is 3
Do you want to continue? (y/n): y
```

```
Main Menu
1. Insert
2. Delete
3. Display
Enter your choice: 3

The elements are:
1 2 3

Do you want to continue? (y/n): n
```